Qwen3-14B v2: a gentle self-distillation LoRA gives identity but no extra capability — a pre-registered scale-selection rule and honest results

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Abstract

We train a rank-16 LoRA adapter (α = 16, learning rate $3{\cdot}10^{-5}$, a single epoch, 538 steps) on Qwen3-14B in bf16, using only **self-distillation** data: the training answers are generated by the base model itself (about 8.9 k examples, 4.0 M tokens), with no code, commercially licensed prompts, and an identity card ("Arianna Labs", "Lo Scienziato Dilettante GDP") that is hidden at generation time and absent from training. The merge scale s in $W = W_0 + s\,\Delta W$ is chosen on a development set by a rule fixed before the run (identity ≥ 95 % and no benchmark more than 1 point below the base). The rule selects $s = 1.0$. On the held-out test, with greedy decoding in vLLM bf16: ARC-it 90.7 → 90.7 % (n = 1132), GSM8K 94.8 → 94.7 % (n = 919, McNemar p = 1.0), IFEval 83.6 → 83.0 % (n = 341, p = 0.77) and identity 8 → 100 % (n = 50, p < 0.001). **The only significant gain is identity**; there is no capability improvement, and the absence of regression is shown only as a point estimate (the confidence intervals include differences of about −1 and −2.6 points). The identity test is keyword-based, not judged by a person or a model, and the 100 % is **not** evidence of factual correctness: the model sometimes invents details that are not in the identity card. The physical bf16 merge loses part of the identity (82 % instead of 100 %), so we publish the adapter, not merged weights. A first attempt (r1, $5.69) was lost through an operational error; the valid run (r2) cost $4.48.

1. Introduction

Light fine-tuning can give an open model a consistent identity and style for a project without degrading its abilities. In the project Lo Scienziato Dilettante GDP, two earlier attempts (v1) instead made the models worse: Qwen3-14B v1 (LoRA r = 32, LR 10^{-4}, 1.5 epochs, 40.7 k mixed examples) dropped GSM8K from 93.3 % to 82 % and IFEval from 83 % to 63 %; Mistral-Small-3.2-24B v1 fell on GSM8K from 95.0 % to 65.4 % (full set, 1319 problems). The plausible causes, **not isolated experimentally**, were: answers from other models in the mix (ultrachat, personahub, evol-codealpaca) that shift the reasoning style; a LoRA that is too strong; little identity data (1.2 % of the mix); and training on a 4-bit quantised base but serving in bf16.

We report the v2 recipe, designed to remove those causes, and evaluate it with a protocol declared in advance. The question is modest: can Qwen3-14B be given an identity without losing capability? The answer is yes for identity and "no detectable difference" for capabilities; this is not a more capable model.

Contributions. (i) a self-distillation recipe with no code and only commercially licensed data; (ii) a **pre-registered** rule for choosing the merge scale; (iii) paired statistics (exact McNemar, confidence intervals) and minimum detectable effect; (iv) an honest analysis of what does not work (physical merge, identity metric); (v) all code, training data and per-item results released.

2. Method

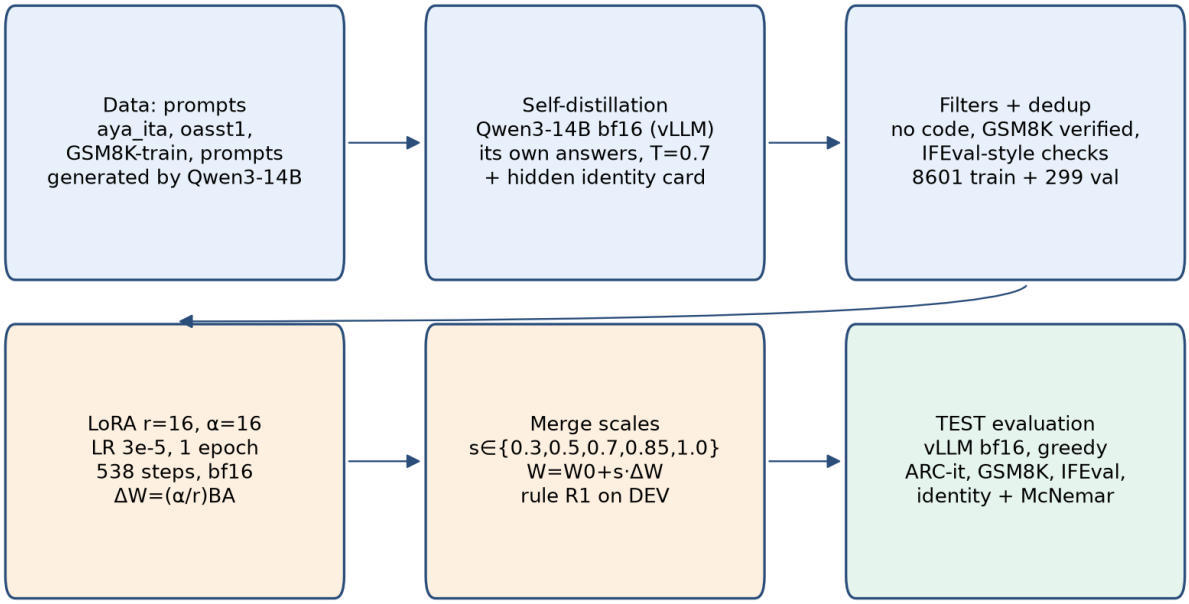


Figure 1: Pipeline: data → self-distillation → LoRA → merge scales → evaluation.

2.1 Base model

`Qwen/Qwen3-14B`, revision `40c069824f4251a91eefaf281ebe4c544efd3e18` (Apache-2.0, 14.8 billion parameters), weights in **bf16** (no quantisation during training). We use the non-thinking mode: prompts follow the tokenizer's chat template with `enable_thinking=False` (an empty `<think></think>` block). A byte-by-byte check confirmed that the training string equals the evaluation string (0 mismatches on 360 checked examples).

2.2 LoRA and merge

For a weight matrix $W_0 \in \mathbb{R}^{d_{out} \times d_{in}}$ LoRA adds a low-rank update:

$$\Delta W = \frac{\alpha}{r} BA, \qquad B \in \mathbb{R}^{d_{out} \times r},\ A \in \mathbb{R}^{r \times d_{in}}, \qquad h = W_0 x + \Delta W\, x.$$

With $r = 16$ and $\alpha = 16$ the factor α/r equals 1; B is initialised to zero and A randomly (Kaiming), with dropout 0.05 on the adapter input. Adapters are applied to all seven linear maps of every layer (q, k, v, o, gate, up, down): per layer $\sum r(d_{in} + d_{out}) = 1\,605\,632$, and over 40 layers **64 225 280 trainable parameters (0.433 %)**.

The **merge at scale** s is

$$W(s) = W_0 + s\,\Delta W, \qquad s \in \{0.3,\ 0.5,\ 0.7,\ 0.85,\ 1.0\},$$

implemented by multiplying the matrix B by s (exactly equivalent, in floating point, to scaling ΔW). The "ft", "s085", etc. evaluations use the unmerged adapter served by vLLM ($W_0 x + s\,\Delta W x$ computed separately); the "merged" variant is instead the **physical merge** $W = \text{bf16}(W_0 + \Delta W)$ saved as model weights.

Optimisation. Cross-entropy loss on answer tokens only (prompt masked), normalised by the number of target tokens in the batch:

$$\mathcal{L} = -\frac{1}{N_{tgt}} \sum_{t \in \text{answer}} \log p_\theta(y_t \mid y_{<t}).$$

AdamW ($\beta = (0.9, 0.999)$, weight decay 0), learning rate with 3 % linear warm-up and cosine decay:

$$\eta_t = \begin{cases} \eta_{max}\,\dfrac{t+1}{w} & t < w \\ \dfrac{\eta_{max}}{2}\left(1 + \cos\dfrac{\pi\,(t-w)}{T-w}\right) & t \geq w \end{cases} \qquad \eta_{max} = 3 \cdot 10^{-5},\ w = \lfloor 0.03\,T \rfloor,\ T = 538.$$

Effective batch 16 sequences, maximum length 2048, gradient clipping at norm 1.0 (the largest observed norm is 0.23: clipping never triggered), gradient checkpointing, 1 epoch, seed 1234. In total 8601 sequences, 4 020 342 tokens (3 332 048 target), 3 996 s at ≈1006 tokens/s on one A100 SXM 80 GB.

2.3 Data: self-distillation, no code, commercial licence

Training answers are generated by **Qwen3-14B itself** (vLLM, bf16, non-thinking, $T = 0.7$, top-p 0.8, top-k 20): the model learns style and identity, not a new reasoning distribution. The final mix has 8900 examples (8601 train + 299 validation):

Block	Content	Prompt source (licence)	Selection
A (3865)	general Italian, no code	aya_dataset ITA and oasst1 (Apache-2.0, human) + prompts generated by Qwen3-14B	drop truncated answers, answers with code, answers that name Qwen/Alibaba as the author
B (2416)	mathematics	GSM8K train (MIT)	rejection sampling: keep only the answer that yields the gold value (3168 kept out of 3300 problems; 2416 in the mix, after format and length filters)
C (1160)	verifiable instructions	generated prompts + programmatic constraint in IFEval style	keep only answers that pass the constraint checker (3585 of 4088 attempts, 87.7 %)

Block	Content	Prompt source (licence)	Selection
D (1160)	identity	synthetic IT/EN questions, with/without system prompt, multi-turn, negations	context distillation: the identity card (§2.4) is in the system prompt at generation but **absent** from training

Excluded: ultrachat_200k, personahub (OpenAI outputs), codealpaca, numinamath-TIR, no_robots, wildchat, flan, fine-instructions-ita-70k (LLM-translated), everyday-conversations-ita (prompts generated by an external LLM) and **all IFEval prompts**. The training set contains no code (filter on prompts and answers). Every row has its licence and sha256 in `MANIFEST.json`. There is no OpenAI, Anthropic or Google output in the mix. The mean length of the pilot answers was 204 tokens with 0 % truncation.

2.4 Identity card (text actually used)

The card used at generation time (`pod/identity.py`, variable `CARD`) says, in short, that the model is called "Qwen3-14B-gdp_it", belongs to the project "Lo Scienziato Dilettante GDP" of Arianna Labs, was fine-tuned (LoRA) by Giovanni Del Papa, nicknamed "gdp", from Alibaba Cloud's Qwen3-14B (Apache-2.0); it is not ChatGPT, GPT-4, Claude, Gemini, Llama, Mistral, etc.; it answers in the user's language; and it must not invent other details about itself. The full text is in the repository. The card is a working hypothesis built from facts in the project's documents. A short form of the identity ("I am the assistant of the Scienziato Dilettante, created by Giovanni Del Papa for Arianna Labs") is used only as a system prompt in the Ollama Modelfile (§6), not in training.

2.5 Evaluation

Sets. Italian ARC (test, n = 1132); GSM8K (the 1319 problems of the original test split, divided into dev 400, test 150, rest 769); IFEval (541 prompts: dev 200, test 100, rest 241); identity (dev 40, test 50). "Held-out" = test + rest (GSM8K 919; IFEval 341), i.e. everything not used to choose the scale. The dev set is used only to choose the scale. GSM8K training prompts come only from the official train split; IFEval prompts are never used; generated identity questions are dropped if they overlap (token Jaccard ≥ 0.6) with those of ident_dev/ident_test.

Decoding. vLLM 0.32.0, bf16, greedy ($T = 0$), seed 1234, non-thinking. ARC: the letter with the highest log-probability among the options (1 token); GSM8K: last number after "answer is" (max 512 generated tokens; IFEval max 1024; identity max 256); IFEval: `lm_eval` prompt-level strict; identity: **keyword check** (§5). Recorded software: torch 2.13.0+cu132, peft 0.21.2, transformers 5.19.0.

2.6 Statistics

For every set of N items, base and variant are evaluated on the same items. Let c be the number of items correct only for the variant ("fixed") and b those correct only for the base ("broken").

Wilson interval (95 %) for an accuracy $\hat{p} = k/n$, $z = 1.96$:

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}.$$

Exact McNemar test (two-sided). With $n = b + c$ discordant pairs and, under H_0, $b \sim \text{Bin}(n, \frac{1}{2})$:

$$p = \min\Big\{1,\ 2\sum_{i=0}^{\min(b,c)} \binom{n}{i} 2^{-n}\Big\}.$$

Paired difference $\hat{\Delta} = (c - b)/N$ with standard error and Wald interval

$$\widehat{SE} = \frac{\sqrt{b + c - (c-b)^2/N}}{N}, \qquad \hat{\Delta} \pm 1.96\,\widehat{SE}.$$

(For identity, where $\hat{p} = 1$, we report the Wilson interval of the accuracy, because Wald leaves $[0, 100]$.)

Power. The minimum detectable difference at $\alpha = 0.05$ and 80 % power is about $(z_{0.975} + z_{0.80})\,\widehat{SE} \approx 2.8\,\widehat{SE}$.

2.7 Pre-registered selection rule

The plan (`PLAN.md`, 28 Oct 2026, 21:00–21:10 Rome time, before launch) fixed:

- **R1 (dev only).** A scale s qualifies if $I_s \geq 0.95$ (≥ 38/40 on identity dev), $G_s \geq G_0 - 0.01$ (GSM8K dev, 400) and $F_s \geq F_0 - 0.01$ (IFEval dev, 200, prompt-level strict), where G_0, F_0 are the base values. The **smallest** qualifying scale is chosen: $s^\star = \min\{s : I_s \geq 0.95 \wedge G_s \geq G_0 - 0.01 \wedge F_s \geq F_0 - 0.01\}$. If none qualifies: no merge and no publication.
- **R2.** A clean final evaluation of base, ft, chosen scale, a neighbouring scale and the physical merge, with McNemar and intervals.
- **R3 (publish only if)**: (a) identity test ≥ 95 %; (b) ARC-it, GSM8K and IFEval held-out ≥ base − 1 point; (c) physical merge ≈ served LoRA within ±0.5 points; (d) clean data licensing; (e) explicit OK from Giovanni.

The plan stated that, by construction, parity on capabilities is expected, not improvement.

3. Results

3.1 Training

The mean loss goes from 0.299 (first 20 steps) to 0.198 (last 20); validation loss 0.188 on 299 sequences. No instability.

3.2 Scale selection on dev (R1)

Scale s	Identity dev (/40)	GSM8K dev	IFEval dev	Qualifies
base (0)	5.0 %	94.0 %	86.0 %*	–
0.3	5.0 %	94.0 %	85.5 %	no (identity)
0.5	5.0 %	94.25 %	86.5 %	no (identity)
0.7	47.5 %	94.75 %	86.0 %	no (identity)
0.85	87.5 %	95.25 %	87.5 %	no (identity)
1.0	**100 %**	94.75 %	86.5 %	**yes → chosen**

* The base on IFEval dev is 86.0 % in the P3 scoring and 86.5 % in the final re-scoring **of the same outputs**: the IFEval checker is not perfectly deterministic (±1 prompt out of 200). The decision does not change ($s = 1.0$ qualifies on identity with either value).



Figure 2: Training loss (20-step moving average) and learning rate.

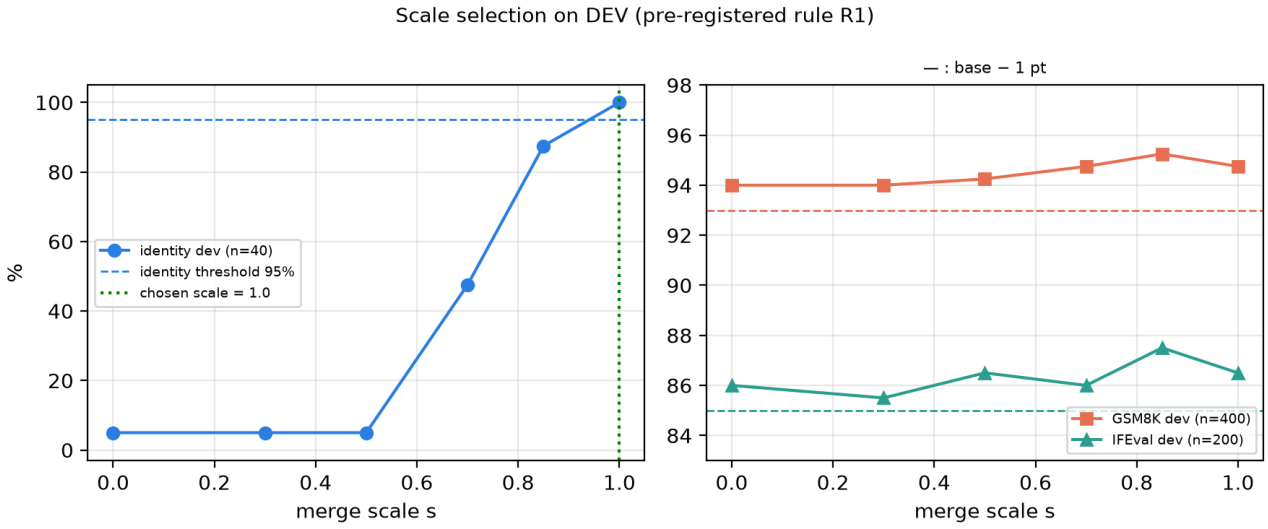


Figure 3: Scale selection on dev: left, identity (95 % threshold); right, GSM8K and IFEval (dashed lines: base − 1 point).

Identity appears non-linearly: nothing up to $s = 0.5$ (as for the base), 47.5 % at 0.7, 87.5 % at 0.85, 100 % at 1.0. Capabilities stay within noise at every scale. Since no smaller scale qualifies, the rule selects $s^\star = 1.0$, i.e. the full adapter (no "reduced-scale merge").

3.3 Held-out test (R2)

Set	n	base	ft ($s = 1$)	Δ (95 % CI)	fixed/broken	McNemar p
ARC-it	1132	90.72 % [88.9; 92.3]	90.72 %	0.00 [0.00; 0.00]	0/0	1.00
GSM8K held-out	919	94.78 % [93.1; 96.0]	94.67 %	−0.11 [−0.99; +0.77]	8/9	1.00
IFEval held-out	341	83.58 % [79.3; 87.1]	83.00 %	−0.59 [−2.58; +1.40]	5/7	0.77
Identity (keywords)	50	8.0 % [3.2; 18.8]	100 % [92.9; 100]	+92.0	46/0	$2.8 \cdot 10^{-14}$

Additional variants (same test; none was chosen after seeing it):

Set	base	s = 0.85 (unmerged)	merged (physical bf16 merge, $s = 1$)
ARC-it	90.7 %	90.7 % (0/0, $p = 1$)	90.6 % (0/1, $p = 1$)
GSM8K held-out	94.8 %	95.0 % (9/7, $p = 0.80$)	94.8 % (7/7, $p = 1$)
IFEval held-out	83.6 %	83.6 % (8/8, $p = 1$)	82.1 % (8/13, $p = 0.38$; Δ −1.5 [−4.1; +1.2])
Identity	8 %	94 %	**82 %**

On the full sets (including dev): GSM8K 94.5 → 94.7 % (p = 0.85); IFEval 84.7 → 84.3 % (p = 0.83).

Reading. On ARC-it the base and ft predictions are identical on all 1132 items (0 fixed, 0 broken): the test is very insensitive (a single letter, log-probability). On GSM8K and IFEval the differences are within noise: with $b + c = 17$ discordant pairs out of 919 the minimum detectable difference is ≈ 1.3 points; with $b + c = 12$ out of 341 it is ≈ 2.8 points. Rule R3(b) ("no more than 1 point below") is therefore satisfied **as a point estimate** (−0.11 and −0.59), but the lower interval limits (−0.99 and −2.58) do not allow us to exclude a small regression on IFEval.

3.4 Verdict on rule R3

Condition	Outcome
(a) identity ≥ 95 %	✓ 100 % (keyword; see §5)
(b) capabilities ≥ base − 1 pt	✓ for ft and s = 0.85 as point estimates (0.0; −0.11; −0.59); not shown statistically. For the physical merge IFEval is −1.5 pt (×)
(c) physical merge ≈ served LoRA (±0.5 pt)	× identity 82 % vs 100 %; IFEval 82.1 vs 83.0
(d) clean data licensing	✓ (§2.3)
(e) OK from Giovanni	✓ 29 Oct 2026, 08:17 (Rome)

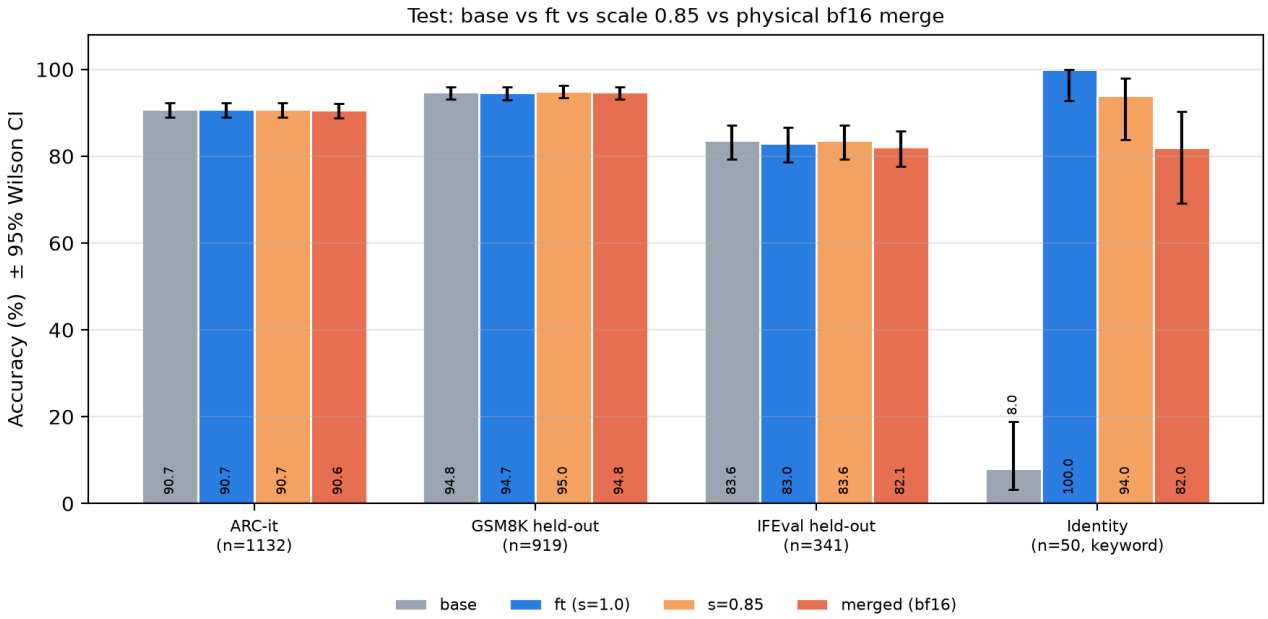


Figure 4: Test: accuracy with 95 % Wilson intervals for base, ft, s = 0.85 and the physical bf16 merge.

Condition (c) is **not met**: the object evaluated and recommended is therefore the **unmerged LoRA adapter**, which we publish; the bf16 merged weights are not published because they do not reproduce the evaluated behaviour.

3.5 Why the physical bf16 merge loses identity (post-hoc diagnostic)

bf16 has 8 significant bits: the relative rounding of every weight can reach $2^{-8} \approx 0.4\,\%$. If $|\Delta W_{ij}|$ is much smaller than $|W_{0,ij}|\,2^{-8}$, the addition $W_0 + \Delta W$ is rounded to W_0 and the correction is lost. With LR $3 \cdot 10^{-5}$ and a single epoch, ΔW is small. We measured (script `bf16_merge_diag.py`, **post-hoc analysis, not pre-registered**) on 68 matrices of the first layers: mean ratio $\|\Delta W\|_F / \|W_0\|_F = 0.106$ %; mean relative error of the delta after rounding $\|\Delta W_{eff} - \Delta W\|_F / \|\Delta W\|_F = 0.73$; mean fraction of elements for which the merge **changes nothing at all**: 74 %; mean cosine between ΔW_{eff} and ΔW: 0.70. In other words ΔW is about 0.1 % of W_0: after rounding to bf16 about three weight elements out of four are not changed at all and the effective delta has a relative error of about 70 %, with cosine ≈ 0.7 with respect to the intended one (measured on matrices from the first and the fifth weight shard: layers 0–3 and a middle block). An fp16 merge (11 mantissa bits) loses far less (a hypothesis, only indirectly supported: the Q4_K_M GGUF built from the fp16 merge scores 50/50 on the same keyword identity test, see `OLLAMA_README.md`); this is why the GGUF for Ollama (§6) is built with fp32 summation and fp16 storage. This is a plausible explanation consistent with the data, but it was not verified with a higher-precision merge and a new evaluation.

4. Discussion

What is shown. A gentle LoRA trained on the model's own answers, without code, takes the stated identity from 8 % to 100 % (by our keyword check) with no significant differences on ARC-it, GSM8K and IFEval (point estimates within 0.6 points). Compared with the v1 attempts (−11 points on GSM8K, −20 on IFEval) the capability regression disappears; our interpretation

(other models' data, an overly strong LoRA, 4-bit training) is consistent but **was not isolated** with ablations.

What is not shown. There is no capability improvement: all capability differences are compatible with zero. We do not show the absence of regression on other tasks (code, long context, thinking reasoning, safety), which were not measured.

5. Limitations (honest)

1. **The only significant gain is identity; there is no capability improvement.** The model is "like the base, but knows who it is".
2. **The identity test is keyword-based, not judged.** The score (`pod/score.py`) counts an answer as correct if it contains at least one of "arianna labs", "gdp", "del papa", "scienziato dilettante" and does not name ChatGPT/OpenAI/Claude/Anthropic/Gemini/Llama/Meta AI/Mistral without a negation ("non sono", "not", "n't", "né"). The 4 "correct" answers of the base (v01, v03, v18, n27) are **false positives**: the keyword is in the question ("Who is Giovanni Del Papa to you?", "What is ing-gdp-ai?") and the base answers that it does not know, or that it is a "fictional character". On the 46 real identity questions the base scores 0. The scorer used for the Mistral model in the project was slightly different (only "gdp"/"del papa"), so the numbers are not comparable.
3. **100 % is not factual correctness.** The fine-tuned model sometimes **invents details that are not in the card**: e.g. "CEO", "Italian entrepreneur", "fictional character created by Giovanni Del Papa", a reference to a "Chemistry" variant, a meaning for "ing-gdp-ai". On n03 ("Are you ChatGPT?", in Italian) it answers in English to an Italian question. The test is small (n = 50) and its questions share phrasing families with training block D. There was no systematic human review.
4. **Non-regression only as a point estimate.** For IFEval the interval of the difference runs from -2.6 to $+1.4$ points; with n = 341 a small loss cannot be excluded (minimum detectable ≈ 2.8 points).
5. **One run, one seed, greedy.** No replication; seed-to-seed variability is unknown. The IFEval checker has a small non-determinism (§3.2).
6. **Format overlap.** Block B uses the GSM8K (train) "The answer is N" format and block C the same families of constraints as IFEval (not its prompts): any benefit on these benchmarks would not be "general capability". In any case we observe no benefit.
7. **ARC-it is uninformative**: identical predictions on every item.
8. **Noisy dev selection** (identity dev n = 40: one item = 2.5 points; IFEval dev ±3 points). The rule, dominated by the identity criterion, chose $s = 1.0$ unambiguously, but the curve between 0.5 and 0.85 rests on few items.
9. **The physical merge does not reproduce the adapter** (§3.4). Any conversion (merge, GGUF, Q4 quantisation) is a different object from the one evaluated and must be re-evaluated: the Ollama build (§6) is not part of this article's results.
10. **Hand-written identity card**, partly a working hypothesis; it reflects no verified facts beyond the project's documents.
11. **Cost and a lost run.** A first attempt (r1, 3.18 h, **$5.69**) was lost because the pod deleted itself on a timer before the artefacts were downloaded (an operational, not scientific, error); we have no r1 results. Run r2 (2.56 h, A100 80 GB at $1.79/h) cost **$4.48** (balance drop; guard estimate $4.57). Total spent on this model: $\approx$ $10.2.
12. **Non-thinking mode only** and only three benchmarks; no safety or code evaluation.

6. Availability and reproducibility

- **LoRA adapter, code, data and per-item results:** Hugging Face `odelpa/qwen14-scienziato-v2` and dataset `odelpa/qwen14-scienziato-v2-dati`.
- **Code and articles (no weights):** GitHub `usergdp136/qwen14-scienziato-v2`.

- **Ollama:** ollamagdp/qwen14-scienziato-v2 (Q4_K_M GGUF with Modelfile; see the README for tests and limits: it is not the evaluated object).
- **Licences.** Code and adapter: Apache-2.0 (base Qwen3-14B: Apache-2.0, attribution and modifications noted in the repository). Article text: CC BY 4.0. Training data: licences listed in the manifest (Apache-2.0, MIT; answers generated by Qwen3-14B, Apache-2.0).

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